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Paper 1 of 1 — CHE 412 PROCESS INSTRUMENTIZATION

Level 400 • First Semester • Handout

CHE 412

PROCESS INSTRUMENTATION

Lecture Material: Ohimor E.O.

Page 1

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

DEPARTMENT OF CHEMICAL ENGINEERING

COURSE CODE: CHE 412

COURSE TITLE: PROCESS INSTRUMENTATION

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1. COURSE BACKGROUND AND PURPOSE

The objective of this course is to introduce the student to the essentials of process instrumentation; principle of operation of process control instruments and areas of application.

2. COURSE OUTCOMES

At the end of the course, the student will be able to:

- i. Demonstrate understanding of measurements, errors etc.
- ii. Demonstrate knowledge of process instruments (sensors, transducers etc) required for efficient and safe operation of industrial process plants.
- iii. Explain how these process instruments are used to sense, measure and transmit process variables information to a control system.
- iv. Appreciate the need for calibration of process instruments and also carry out same.
- v. Describe the various process variables (flow, level, pressure, temperature, analytical etc) found in a process plant.

3. COURSE CONTENT

This course consists of:

- Basic elements of process control:
- The process sensors, final control elements, the controller.
- Instrumentation Basics: Sensors, transducers instrumentation signals, transmitters, instrumentation dynamics, instrumentation noise.
- Process measurements and instruments: Temperature measurement, pressure measurement, flow measurement, valves and pumps.
- Process characteristics: First order process, higher order processes, process type

4. PRIOR KNOWLEDGE ASSUMED

Students are assumed to have basic knowledge of various process variables; flow, level, pressure, temperature in other lower-level course such as Introduction to Oil and Gas

Operations (PEE 211).

5. ASSESSMENT

Mid-semester class test- 30 %

Final examination - 70 %

Lecture Material: Ohimor E.O.

Page 2

6. TEACHING AND LEARNING PROCESS

The course is almost entirely based on classroom discussions, review of articles on Process Instrumentation. It will involve class tests and the final exams, which will be conducted as specified in the University Examination Timetable.

7. INFORMATION TO SUPPORT THE COURSE

The prescribed text is:

- i. Process Instrumentation, Dynamics and Control for Chemical Engineers by Otoikhian Shegun Kevin, Ohikhen Sule Olohi, Olayebi Oluwafemi, and Otu Francis Ibiefo. 1st Edition.
- ii. Measurement and Instrumentation Principles by Allan S, Morris. 3rd Edition.

Lecture Material: Ohimor E.O.

Page 3

CHE 412 PROCESS INSTRUMENTATION

TABLE OF CONTENTS

Lecture Material: Ohimor E.O.

Page 4

CHAPTER 1

BASIC CONCEPTS AND QUALITIES OF MEASUREMENT

1.0 INTRODUCTION

Measurement systems are traditionally used to measure physical and electrical quantities, such as mass, temperature, force, pressure, velocity, angular velocity, acceleration, capacitance, and voltage.

However, they also can be designed to locate things or events, such as the epicenter of an earthquake, employees in a building, partial discharges (PDs) in a high-voltage (HV) power cable, or a land mine.

Often a measurement system is called upon to single out and count objects, such as red blood cells, fish of a certain size swimming past a check point, or bottles of beer on a conveyor.

Also, a measurement system may have the task of measuring the concentrations of chemical substances, such as various ions in solution such as H^+ (pH), Na^+ , K^+ , Cl^- , or SO_4^{2-} , the concentrations of antibodies or other biological molecules in solution, or the concentrations of certain explosive or toxic gasses such as CH_4 or carbon monoxide.

A measurement system is often made part of a control or regulatory system. The old saying

that “if you can’t measure it, you can’t control it” is certainly a valid axiom for the control engineer as well as for the instrumentation engineer.

Thus, for the control engineer as well as for the instrumentation engineer it is important to also know how noise, calibration errors, sensor dynamic response, and nonlinearity can affect the accuracy, precision, and resolution of measurements. Figure 1.0 shows the basic elements of a measuring instrument.

Figure 1.0 Basic elements of a measuring instrument

Lecture Material: Ohimor E.O.

Page 5

1.1 Measurement System Architecture

Figure 1.1 illustrates the block diagram of a typical measurement system. The quantity under measurement (QUM) is converted to a useful form, such as a voltage, current, or physical displacement by an input transducer or sensor.

The distinction between a sensor and a transducer: both devices are sensors (signal collectors); however, transducers have the property of being able to convert an input signal to some analog quantity.

An example of a transducer is a piezoelectric crystal. A mechanical displacement (or, equivalently, a force or pressure) applied to the crystal produces an output voltage; an input voltage to the crystal produces a mechanical displacement. Another example of a transducer is a loudspeaker: a current through the voice coil causes cone displacement; the rate of mechanical displacement generates a proportional current output because the voice coil moves in a magnetic field.

FIGURE 1.1 A generalized measurement system. Note: QUM = x = quantity under measurement. S =

sensor. A = amplifier. SCF = signal conditioning filter (analog). DSO = digital storage oscilloscope. ATR =

analog tape recorder. AAF = anti-aliasing filter (analog). n_1 = noise accompanying the QUM. n_2 = equivalent noise from electronics. n_3 = equivalent quantization noise. ADC = analog-to-digital converter.

The digital computer can further filter and store digitally V_o , the conditioned sensor output: $V_o = x + n_1 + n_2$

The output of the sensor, here assumed to be a voltage, is amplified and filtered by a signal conditioning subsystem. The purposes of the signal conditioning subsystem are to amplify, Lecture Material: Ohimor E.O.

Page 6

give a low or matched output impedance, and improve the signal-to-noise ratio (SNR) of the analog signal proportional to the QUM. The conditioned analog signal, V_o , can be distributed to various display and recording devices. V_o can be displayed on an analog or digital oscilloscope or a strip-chart recorder. It can also be recorded on a magnetic tape recorder, or it can be low-

pass filtered to prevent aliasing and then periodically converted to digital words by an analog-to-digital converter (ADC). As you will see, the sensor output can be wirelessly communicated (from sensor S to amplifier A) to the rest of the measurement system by modulated radio frequency or light beam.

The ADC output, N , is generally made an input to a digital computer through an appropriate A/D interface. Once in the computer environment, the sampled and digitized QUM can be further (digitally) filtered and processed and stored on a magnetic hard and/or floppy disk or on an optical disk in digital form. Often the sensing instrument contains an ADC and is connected to a computer by a special instrumentation bus such as the USB (universal serial bus), GPIB (general purpose interface bus), or HPIB (Hewlett-packard interface bus). The General Purpose Interface Bus (GPIB) is organized so that many GPIB-compatible sensing instruments can be managed by one master (host) computer.

Note that a bus (binary unit system) is a subsystem that transfers data between computer components inside a computer or between computers.

There are three types of bus:

Address bus – carries memory addresses from the processor to other components such as primary storage and input/output devices

Data bus – carries the data between the processor and the components

Control – carries control signals from the processors to other components.

Note that there are three major sources of noise in the typical measurement system shown in Figure 1.1:

- noise sensed along with the QUM (environmental noise),
- noise associated with the electronic signal conditioning system (signal or input conditioning noise), and
- equivalent noise generated in the analog-to-digital conversion process (quantization noise).

These noise sources generally provide a limit to measurement system resolution and to its accuracy. Many sensors provide the rate-limiting element in a measurement system. That is, the response speed of the sensor is much slower than any of the other elements in the system. As a result of sensor dynamics, a measurement system can require a minimum settling time before a stable measurement can be made when there is a step change in the QUM. That is, a settling time is required for the sensor to reach a steady-state (SS) output.

Lecture Material: Ohimor E.O.

Page 7

1.2.1 Sensor Dynamics

Sensor dynamics generally fall into one of three categories: first-order low-pass, second-order low-pass, and band-pass. A general example of a first-order, low-pass sensor is shown in Figure 1.2. Here, the QUM is $x(t)$, and the sensor output is a voltage, $v_x(t)$. The ordinary differential equation (ODE) relating v_x and x is

FIGURE 1.2 Transfer function of a sensor with first-order, low-pass dynamics in Laplace format. $x(t)$ is

the QUM; $X(s)$ is its Laplace transform. $v_x(t)$ is the sensor's output voltage, $V_x(s)$ is its Laplace transform,

where a is the natural frequency in r/s of the sensor

1.3 ERRORS IN MEASUREMENTS

Errors in measurements can arise from many causes; there are remedies for some types of errors, but others haunt us as intrinsic properties of the measurement system under use, and often can be mitigated by system redesign.

A. Gross errors in a measurement that can arise from such human mistakes as follows:

1. Reading the instrument before it has reached its steady-state. Such a premature reading produces a dynamic error.
2. Not eliminating parallax when reading an analog meter scale and incorrect interpolation between analog meter scale markings.
3. Mistakes in recording measured data and in calculating a derived measurand.
4. Misuse of the instrument. A simple example of this source of error is when a 10 V full-scale analog voltmeter of typical sensitivity (20,000 Ω/V) is connected to a voltage source having a high Thevenin resistance (e.g., 100,000 Ω). Significant voltage is dropped across the internal (Thevenin) resistor.

B. System errors can arise from such factors as follows:

1. The instrument is not calibrated and has an offset; that is, its sensitivity is off, and it needs zeroing. Loss of calibration and zero can occur because of long-term component value changes with aging or changes associated with temperature rise.
2. Reading uncertainty from the presence of random noise. This noise can accompany the measurand and can arise from the signal conditioning electronics in the system.
 - a. External noise from outside is called environmental noise and often can be reduced by appropriate electric and magnetic shielding and proper grounding and guarding practices. Environmental noise is often coherent in nature and can come from poorly shielded radio-frequency sources, such as computers or radio

Lecture Material: Ohimor E.O.

Page 8

stations. Powerline frequency electric and magnetic fields can also be troublesome, as can be ignition noise.

- b. The significant, internally generated random noise may be shown to arise in the first stage of an instrument's signal conditioning amplifier. Some of this noise comes from resistors (Johnson or thermal noise), some comes from the active elements (transistors) in the headstage, and some comes from the quantization or roundoff inherent in the operation of ADCs in modern digital instruments.

3. Slow or long-term drift in the system can destroy the certainty of static measurements, that is, measurements of measurands (quantities intended to be measured), which are not

varying in time. Drifts

can cause slow changes of system sensitivity and/or zero. Drift may arise as the result of a slow temperature change as a system warms up. Many parameter values can change with temperature, such as capacitance, resistance, inductance, the electromotive force (EMF) of standard cells, and the avalanche voltage of pn junctions. Humidity changes can also affect circuit parameter values, including resistance and capacitance. Good system design involves temperature regulation of the system and the use of low or zero temperature coefficient, sealed components.

Drift or system offset can also arise from DC static charges affecting analog electronic circuits. In some cases, the face of an analog meter can become charged with static electricity, which attracts or repels the meter pointer, causing an error.

1.3.1 Concept of Accuracy, Precision and Resolution

The concepts of accuracy, precision, resolution, limiting error (LE), and various error statistics is better explained by considering the quantitative aspects of errors in measurements. Thus, we define the error in the n th measurement as

where

Y_n is the actual, true, defined, or calculated value of the QUM

X_n is the n th measured value of the QUM

Where Y_n may be defined by a high-resolution, primary standard. Nevertheless, error as given in the equation above is generally an accepted definition. In some cases, the absolute value signs in the $\% \epsilon$ expression may be omitted, and the $\%$ error can be negative.

Lecture Material: Ohimor E.O.

Page 9

The accuracy, A_n , of the n th measurement is defined as

The percent accuracy is simply

The precision of the n th measurement is defined as

The resolution in measuring the QUM is related to the precision of the measurement and is basically the smallest unit of the QUM, which can be reliably detected. Resolution can be expressed statistically.

Accurate measurements require the use of a precision measurement system, which is calibrated against a certified, accurate standard.

Other statistics used to describe the quality of a series of measurements are the deviation, d_n ; the average deviation, D_N ; and the standard deviation, S_N , and variance (square of standard deviation) defined as follows:

Lecture Material: Ohimor E.O.

Page 10

1.4 STANDARDS USED IN MEASUREMENTS

1.4.1 Calibration and Standards

One concern of everyone who has occasion to make a measurement is, is the instrument calibrated? As we have seen earlier, calibration is necessary, along with precision, to enable accurate measurements to be made. Calibration implies observing the instrument's performance when measuring an appropriate standard.

A standard is a physical representation of the QUM whose true value is known with great accuracy. Standards can be classified as

1. International standards
2. Primary standards
3. Secondary standards
4. Working standards

International standards are defined by international agreement and are kept at the International Bureau of Weights and Measures (BIPM) in Sèvres, France. An example of an international standard is the kilogram mass. (More will be said about the mass standard in the following.) International standards are not available on a daily basis for calibration or comparison.

Primary standards are maintained in national standards laboratories in countries around the world. Primary standards, representing some of the fundamental physical and electrical units, as well as some derived quantities, are independently measured and calibrated at the various national laboratories and compared against each other. This process leads to grand or world average figures for the standards. Primary standards are used continually, but generally do not leave the national standards labs.

Secondary standards are reference standards that are initially calibrated from primary standards and then used in industry and research labs on a daily basis to calibrate their working standards.

Working standards which are derived from secondary standards are used on a daily basis to check and calibrate working laboratory instruments.

Lecture Material: Ohimor E.O.

Page 11

1.4.2 Units of Measurements

Instruments are designed to measure in either base unit or derived units and are calibrated with the standards of these units:

Table 1.1 SI Base Units

Table 1.2 Derived Units

Lecture Material: Ohimor E.O.

Page 12

CHAPTER 2

PROCESS MEASUREMENT INSTRUMENTS

2.0 INTRODUCTION

A measurement system, which could be referred to as an instrument, a scale, a rule or analyser, is a device meant to facilitate the comparison of an unknown value with a known value as referred to as standard.

Therefore, measurement is a comparison of a given unknown quantity with one of its predetermined standard values adopted as a unit. The result of any measurement is a concrete number consisting of a unit of measurement having its particular name and the number which shows how many times this particular unit is contained in the quantity being measured.

The primary purpose of measurement in process industries is to aid in the economics of industrial operations by improving the quality of the product and efficiency of production.

2.1 FLOW MEASUREMENT

No industrial measurement is more important than the accurate detection of the flow rates of gases, liquids, and solids. An overview is given of the availability and characteristics of some of the most widely used flow sensors.

2.1.1 Differential Pressure (Obstruction-Type) Meters

Differential pressure meters involve the insertion of some device into a fluid carrying pipe that causes an obstruction and creates a pressure difference on either side of the device. Such meters are sometimes known as obstruction-type meters or flow-restriction meters. Devices used to obstruct the flow include the orifice plate, the venturi tube, and the flow nozzle and the Dall flow tube.

When such a restriction is placed in a pipe, the velocity of the fluid through the restriction increases and the pressure decreases; the volumetric flow rate is then proportional to the square root of the pressure difference across the obstruction. The manner in which this pressure difference is measured is important. Measuring the two pressures with different instruments and calculating the difference between the two measurements is not satisfactory because of the large measurement error which can arise when the pressure difference is small. Therefore, the normal procedure is to use a differential pressure transducer, which is commonly a diaphragm type.

Flow-restriction type instruments are popular because they have no moving parts and therefore robust, reliable and easy to maintain. One disadvantage of this method is that the obstruction causes a permanent loss of pressure in the flowing fluid. The magnitude and hence

Lecture Material: Ohimor E.O.

Page 13

importance of this loss depends on the type of obstruction element used, but where the pressure loss is large, it is sometimes necessary to recover the lost pressure by an auxiliary pump further down the flow line. This class of device is not normally suitable for measuring the flow of slurries as the tapings to the pipe to measure the differential pressure are prone to blockage, although the venturi tube can be used to measure the flow of dilute slurries.

Figure 2.1 obstruction devices: (a) orifice plate; (b) venturi; (c) flow nozzle; (d) Dall flow tube

(a) Venturi and Identical Flowmeter

A number of obstruction devices are available that are specially designed to minimize the pressure loss in the measured fluid. These have various names such as venturi, flow nozzle and dall flow tube. They are all much more experience than an orifice plate but has better performance. The smooth internal shape also means that they suffer much less wear, and consequently have a longer life span than orifice plates. They also require less maintenance and give greater measurement accuracy.

Venturi Meter

The venturi tubes, shown in Figure 5.2(a) uses, the same differential pressure principle as the orifice plate. The venture tube normally uses a specific reduction in tube size, and is normally well suited for use in larger diameter pipes, but it becomes heavy and excessively long. One advantage of the venturi tube is its ability to handle large amounts of suspended solids. It creates less turbulence and insertion loss than the orifice plate. The differential pressure taps in the venture tube are located at the minimum and maximum pipe diameters.

Lecture Material: Ohimor E.O.

Page 14

Figure 2.2 Types of Constrictions used to Measure Flow: (a): Venturi Tube, (b): Flow Nozzle.

The venturi has a precision engineered tube of a special shape. This offers measurement uncertainty of only $\pm 1\%$. However, the complex machining required in manufacturing it means that it is the most expensive of all the obstruction devices discussed. Permanent pressure loss in the measured system is 10-15% of the pressure difference ($P_1 - P_2$) across it.

(b) Flow Nozzle

Nozzles are used to determine fluid flow rate through a pipe. The flow nozzle is a good compromise on cost and accuracy between the orifice plate and the Venturi tube for clean liquids. It is not normally used with suspended particles. Its main use is the measurement of steam flow. The flow nozzle is shown in Figure 2.2b. The pressure loss imposed on the flowing fluid is 30-50% of the measured pressure difference ($P_1 - P_2$) or differential pressure. Flow nozzles are also used as measuring elements for air and gas flow in industrial application.

(c) Orifice Plate

The orifice plate is normally a simple metal diaphragm with a constricting hole. The diaphragm is normally clamped between pipe flanges to give easy access. The differential pressure ports can be located in the flange on either side of the orifice plate, or alternatively, at specific locations in the pipe on either side of the flange as determined by the flow patterns (named vena contracta), as shown in Figure 2.3. Shown also is pressure profile. A differential pressure gauge is used to measure the difference in pressure between the two ports. The differential pressure gauge can be calibrated in flow rates. The lagging edge of the hole in the diaphragm is beveled to minimize turbulence. In fluids, the hole is normally centered in the diagrams, as shown in Figure 2.4(a).

However, if the fluid contains particulates, the hole could be placed at the bottom of the pipe, as shown in figure (b), to prevent a build up of particulates. The hole also can be in the form of

a semicircle having the same diameter as the pipe, and located at the bottom of the pipe, as shown in figure 2.4 (c).

Vena contracta is the point in a fluid stream where the diameter of the stream is the least, and fluid velocity is at its maximum. It can also be described as a point with the least cross-sectional area of fluid stream.

Lecture Material: Ohimor E.O.

Page 15

Figure 2.3 Orifice Constriction Plate with Pressure Profile.

Figure 2.4 Orifice shapes and locations used (a) with fluids, and (b, c) with Suspended Solids.

Orifice plates are simple, cheap and available in a wide range of sizes. In consequence, they account for almost 50% of the instruments used in industry for measuring volume flow rate.

One limitation of the orifice plate is that its inaccuracy is typically at least $\pm 2\%$ and may approach $\pm 5\%$. Also, the permanent pressure loss caused in the measured fluid flow is between 50% and 90% of the magnitude of the pressure difference ($P_1 - P_2$). Other problems with the orifice plate are a gradual change in the discharge coefficient over a period of time as the sharp edges of the hole wear away, and a tendency for any particle in the flowing fluid to stick behind the hole and thereby gradually reduce its diameter as the particles build up. The latter problem can be minimized by using an orifice plate with an eccentric hole. If this hole is close to the bottom of the pipe, solids in the flowing fluid tend to be swept through, and build up of particles behind the plate is minimized. A very similar problem arises if there is any bubble of vapour or gas in the flowing fluid when liquid flow is involved. These also tend to

Lecture Material: Ohimor E.O.

Page 16

build up behind an orifice plate and distort the pattern of flow. This difficulty can be avoided by mounting the orifice plate in a vertical run of pipe.

The flow rate Q in a differential flow rate meter is given by:

Where K is the flow coefficient constant, d_o is the diameter of the orifice, d_p is the pipe diameter, and h is the difference in height between P_H and P_L .

Example 5.1

In a 0.76m diameter pipe, a circular orifice has a diameter of 0.51m, and the difference in height of the manometer levels is 0.70m. What is the flow rate in cubic meters per second, if K is 0.97?

(d) Pitot Static Tube

The pitot static tube, as shown in Figure 2.5 is an alternative method of measuring flow rate. The pitot static tube is mainly used for making temporary measurements of flow, although it is also used in some instances for permanent flow monitoring. It measures the local velocity of flow at a particular point within a pipe rather than the average flow velocity as measured by other types of flowmeter. This may be very useful where there is a requirement to measure local flow rates across the cross-section of a pipe in the case of non-uniform flow. Multiple

Lecture Material: Ohimor E.O.

Page 17

pitot tubes are normally used to achieve this. Thus, a pitot-static tube is capable of measuring both stagnation pressure and static pressure in the pipe.

The differential or dynamic pressures in any of the above devices can therefore, be measured using the pressure measuring sensors.

In a pitot static tube, the flow Q is given by:

Where K is a constant, P_s is the static pressure, and P_t is the impact pressure.

Figure 2.5: Pitot Static Tube

The instrument depends on the principle that a tube placed with its open end in a stream of fluid, as shown in figure 2.6, will bring to rest that part of the fluid which impinges on it and the loss of kinetic energy will be converted to a measurable increase in pressure inside the tube. The static pressure (P_2) is measured. The flow velocity can then be calculated from the formula:

Lecture Material: Ohimor E.O.

Page 18

The constant C , known as the Pitot tube coefficient, is a factor which corrects for the fact that not all fluid incident on the end of the tube will be brought to rest; a proportion will slip around it according to the design of the tube. Having calculated V , the volume flow rate can then be calculated by multiplying V by the cross-sectional area of the flow pipe, A . Pitot tubes have the advantage that they cause negligible pressure loss in the flow.

Figure 2.6: Pitot Static tube

They are also cheap, and the installation procedure consists of the very simple process of pushing them down a small hole drilled in the flow-carrying pipe. Their main failing is that the measurement inaccuracy is typically about $\pm 5\%$, although more expensive versions can reduce inaccuracy down to $\pm 1\%$. The annular is a development of the pitot tube that has multiple sensing ports distributed across the cross-section of the pipe. It thus provides only an approximate measurement of the mean flow rate across the pipe.

(e) Pitot Tube

A pitot tube is a pressure measurement instrument used to measure fluid flow velocity. The pitot tube is used to measure the local velocity at a given point in the flow stream and not the average velocity in the pipe or conduit. A pitot tube, only measures stagnation pressure at the nose of the probe.

Theory of Operation

The basic pitot tube consists of a tube pointing directly into the fluid flow. As this tube contains fluid, a pressure can be measured; the moving fluid is brought to rest (stagnated) as there is no outlet to allow flow to continue. This pressure is the stagnation pressure or the pitot pressure, also known as the total pressure. The measured stagnation pressure cannot itself be used to

determine the fluid velocity. However, Bernoulli's equation states:

Lecture Material: Ohimor E.O.

Page 19

Stagnation pressure = static pressure + dynamic pressure.

This can also be written as:

Solving the above equation for velocity we get

Where V is fluid velocity; P_t is stagnation or total pressure; P_s is static pressure; and ρ is fluid density.

It should be noted that the above equation applies only to fluids that can be treated as incompressible. Liquids are treated as incompressible under almost all conditions. Gases under certain conditions can be approximated as incompressible.

The value for the pressure drop $P_2 - P_1$, or ΔP due to Δh is the reading on the manometer

Where ρ is the density of the fluid in the manometer and Δh is the manometer reading.

The dynamic pressure is the difference between the stagnation pressure and the static pressure. The static pressure is generally measured using the static ports. The dynamic pressure is then determined using a diaphragm inside an enclosed container. If the air on one side of the diaphragm is at the static pressure, and the other at the stagnation pressure, then the deflection of the diaphragm is proportional to the dynamic pressure, which can then be used to determine the indicated fluid velocity. The diaphragm arrangement is typically

Lecture Material: Ohimor E.O.

Page 20

contained within the fluid velocity indicator, which converts the dynamic pressure to fluid velocity reading by means of mechanical levers.

Instead of separate pitot and static ports, a pitot-static tube (also called a Prandtl tube) may be employed, which has a second tube coaxial with the pitot tube with holes on the pitot tube with holes on the sides, outside the direct airflow, to measure the static pressure.

Industrial Applications of Pitot Tube

Pitot tube is widely used to determine the airspeed of an aircraft and to measure air and gas velocities in industrial applications. In industry, the velocities being measured are often those flowing in ducts and tubing where measurements by an anemometer would be difficult to obtain. In these kinds of measurements, the most practical instrument to use is the pitot tube. The pitot tube can be inserted through a small hole in the duct with the pitot connected to a U-tube water gauge or some other differential pressure devices for determining the velocity inside the ducted wind tunnel. One use of this technique is to determine the volume of air that is being delivered to a conditioned space. The fluid flow in a duct can be estimated from

Volume flow rate (m^3/s) = duct area (m^2) x velocity (m/s)

Selection

The most commonly used general-purpose devices are the pressure differential sensors used

with pipe constrictions. The devices will give accuracy in the 3% range when used with solid state pressure sensors, which convert the readings directly into electrical units, or the rotameter for direct visual reading. The venturi tube has the highest accuracy and least energy loss, followed by the flow nozzle, then the orifice plate. For cost effectiveness, the devices are in the reverse order, if large amounts of particulates are present; the venturi tube has the highest accuracy and least energy loss, followed by the flow nozzle, then the orifice plate. The differential pressure devices operate best between 30% and 100% of the flow range.

Lecture Material: Ohimor E.O.

Page 21

Table 2.1 Summary of Flow Meter Characteristics

Lecture Material: Ohimor E.O.

Page 22

2.1.2 Positive Displacement Flow Meters

A positive displacement flow meter accumulates a fixed volume of fluid and then count the number of times the volume is filled to measure flow. Both gas and liquid can be measured in volumetric (litres per second) or mass (kilogram per second) flow rates: If the density of a liquid is known, these measurements can be converted between one another.

A bucket and a stopwatch can be used in describing the operation of a positive displacement meter. The stopwatch is started when the flow starts, and stopped when the bucket reaches its limit. The volume divided by the time gives the flow rate. For continuous measurement, we need a system of continually, filling and emptying buckets to divide the flow without letting it out of the pipe. These continuously filling and emptying volumes amount to the positive displacement flow measurement.

Positive displacement flow meters account for nearly 10% of the total number of flow meters used in the industry and are used in large numbers for metering domestic gas and water consumption. Accuracy can be as low as + 0.5% depending on the cost.

Examples of positive displacement flow meters are rotary piston meter, turbine meter, rotary vane, nutating disk, Impeller meter, bellows meter, gas meter etc.

(a) Rotary piston meter

The rotary piston meter, also known as the reciprocating piston is a common type of positive displacement meter as shown in Figure 2.7. It consists of a slotted cylindrical working chamber that has an inlet port and an outlet port. The piston moves round the chamber such that its outer surface maintains contact with the inner surface of the chamber, and as this happens, the piston slot slides up and down a fixed division plate in the chamber. At the start of each piston motion cycle, liquid is admitted to volume B from the inlet port. The fluid pressure causes the piston to start to rotate around the chamber, and as this happens, liquid in volume C starts to flow out of the outlet port, and also liquid starts to flow from the inlet port into volume A. As the piston rotates further, volume B becomes shut off from the inlet ports, whilst liquid continues to be admitted into A and pushed out of C. when the piston reaches the end point of its

motion cycle, the outlet port is opened to volume B, and the liquid which has been transported round

inside the piston is expelled. After this, the piston pivots about the contact point between the top of its

slot and the division plate, and volume A effectively becomes volume C ready for the start of the next

motion cycle. A peg on top of the piston causes a reciprocating motion of a lever attached to it. This is

made to operate a counter, and the flow rate is therefore determined from the count in unit time multiplied by the quantity (fixed) of liquid transferred between the inlet and outlet ports for each motion cycle.

Lecture Material: Ohimor E.O.

Page 23

Figure 2.7 Rotary Piston Meter

(b) Nutating disk

Nutating disk meter are in the form of a disk that oscillates (sway), allowing a known volume of fluid to pass with each oscillation. The meter is illustrated in Figure 2.8. liquid enters and fills the left chamber. Because the disk is off center, the liquid pressure causes the disk to wobble.

This action empties the volume of liquid from the left chamber to the right chamber, the left chamber is then refilled and the liquid in the right chamber exits. The oscillations of the disk are counted and the total volume measured. This meter is not suitable for measuring slurries. The meter is accurate and expensive, but a low-cost version is available, which is used in domestic water metering and industrial liquid metering.

Lecture Material: Ohimor E.O.

Page 24

Figure 2.8 Nutating Disk Flow Meter

(c) Impeller meter

An impeller flow meter or rotating impeller is a velocity type meter which has an impeller which rotates horizontally on a vertical shaft. The impellers are of different types and some of these impellers are single jet meter; paddle wheel meter and multiple jet meter.

The impeller element is in a housing in which multiple inlet ports direct the fluid flow at the impeller causing it to rotate in a specific direction in proportion to the flow velocity. This meter works mechanically much like a single jet meter except that the ports direct the flow at the impeller equally from several points around the circumference of the element, not just one point, this minimizes uneven wear on the impeller and its shaft rotating impeller are uniformly divided. Velocity meters, normally used to measure flow rate, also can be set up to measure total flow. Multiplying the velocity by the cross-sectional area of the meter can measure total flow.

(d) Turbine meter

A turbine flowmeter consists of a multi-bladed wheel mounted in a pipe along an axis parallel

to the direction of fluid flow in the pipe, as shown in Figure 2.9. The flow of fluid past the wheel causes it to rotate at a rate that is proportional to the volume flow rate of the fluid. This rate of rotation has traditionally been measured by constructing the flowmeter such that it behaves as a variable reluctance tachogenerator.

Lecture Material: Ohimor E.O.

Page 25

This is achieved by fabricating the turbine blades from a ferromagnetic material and placing a permanent magnet and coil inside the meter housing. A voltage pulse is induced in the coil as each blade on the turbine wheel moves past it, and if these pulses are measured by a pulse counter, the pulse frequency and hence flow rate can be deduced. In recent instruments, fibre optics is also now sometimes used to count the rotations by detecting reflections of the tip of the turbine blades. Provided that the turbine wheel is mounted in low friction bearings measurement inaccuracy can be as low as $\pm 0.2\%$. However, turbine flow meters are less rugged and reliable than flow-restriction type instruments, suitable for pure liquid and are badly affected by any particulate matter in the flowing fluid. Turbine meters are smaller and lighter than the restriction type instruments and are preferred for low-viscosity, high-flow measurements.

Figure 2.9 Turbine Flowmeter

(e) Gas meter

A gas meter is a specialized flow meter, used to measure the volume of fuel gases such as natural gas and propane. Gas meters are used at residential, commercial and industrial buildings that consume fuel gas supplied by a gas utility. Gases are more difficult to measure than liquids, as measured volumes are highly affected by temperature and pressure: Gas meters measure defined volume, regardless of the pressurized quantity or quality of the gas flowing through the meter. Temperature, pressure and heating value compensation must be made to measure actual amount and value of gas moving through a meter. The choice of gas

Lecture Material: Ohimor E.O.

Page 26

meter design, depends on the volumetric flow rate of gas to be measured, the range of flows anticipated, the type of gas being measured and other factors.

(f) Bellow meter

Bellow meters are also known as diaphragm type gas meters are positive displacement meters. The meter has two or more chambers formed by movable diaphragms. Gas flow is directed by internal valves, the chambers alternately fill and expel gas, producing a near continuous flow through the meter. As the diaphragms expand and contract, levers connected to cranks convert the linear motion of the diaphragms into rotary motion of a crank shaft which serves as the primary flow element. The shaft can drive an odometer-like counter mechanism or it can produce electrical pulses for a flow computer.

(g) Rotary vane

A rotary vane meter is an example of a positive displacement meter, having a rotor vane or a moving vane fitted in a pipe configuration or an open channel flow. The vane can be spring loaded and have the ability to pivot. By measuring the angle of tilt, the flow rate can be determined. A rotor is positioned eccentrically in a cylindrical housing so that it is almost touching the cylinder. When the rotor starts turning, centrifugal force moves the blades out of the rotor slots and they slide against the internal surface of the cylinder. A cell is formed between two blades creating a volume that changes constantly during rotation. Air enters from the inlet port into a cell when the cell has achieved its maximum air volume. As the cell moves away from the port, its volume becomes smaller and smaller, the air is compressed and the pressure rises. This continues until the pressure in the cell exits through the outlet port.

2.2 TEMPERATURE MEASUREMENT

Temperature is without doubt the most widely measured variable. In the process control of chemical reactions, temperature control is of major importance, since chemical reactions are temperature-dependent. For accurate temperature control, precise measurement of temperature is required. There is need to first consider various temperature scale used, their relation to each other, before discussing the various temperature measurement instrument. There are four temperature scales namely, Fahrenheit (°F), Celsius (°C), Kelvin (K), and the Rankine scale (°R). Conversion between the scales or units is achieved by the factors shown below.

Lecture Material: Ohimor E.O.

Page 27

Devices used for measuring temperature include the following Liquid in glass thermometers, bimetallic thermometers, Vapour Pressure Thermometer, Thermocouples, Optical Pyrometer and Radiation Pyrometer.

(a) Liquid-In-Glass Thermometers

The liquid-in-glass thermometer is a temperature measuring instrument which consists of either mercury or coloured alcohol, contained within a bulb and capillary tube, as shown in Figure 2.10. As the temperature rises, the fluid expands along the capillary tube and the meniscus level is read against a calibrated scale marked on the tube. It has an inaccuracy of only $\pm 0.15\%$. The operating range of the mercury thermometer is from -350°C to $+450^{\circ}\text{C}$.

Figure 2.10 Liquid in glass thermometer

Lecture Material: Ohimor E.O.

Page 28

(b) Bimetallic Thermometers

The bimetallic principle is based on the fact that if two strips of different metals (e.g. nickel-steel alloy) are bonded together, any temperature change will cause the strip to bend, as this is the only way in which the differing rates of change of length of each metal in the bonded strip can be accommodated. In the bimetallic thermostat, this is used as a switch in control applications, if the magnitude of the bending is measured; the bimetallic device becomes a thermometer. For such

purposes, the strip is often arranged in a spiral or helical configuration, as shown in Figure 2.11 and 2.12.

Figure 2.11 Thermal expansion device: bimetallic thermometer

Figure 2.12 (a) Effect of temperature change on a bimetallic strip, and (b) Bimetallic thermometer using a helical bimetallic coil

The device is normally just used as a temperature indicator, where the end of the strip is made to turn a pointer that moves against a calibrated scale. However, some versions produce an electrical output, using either a linear variable differential transformer (LVDT) or a fibre-optic shutter sensor to transduce the output displacement. Bimetallic thermometers are used to measure temperature between -750°C and $+1500^{\circ}\text{C}$. The inaccuracy of the best instruments can be as low as $\pm 0.5\%$ but such devices are quite expensive.

Lecture Material: Ohimor E.O.

Page 29

Thermostat

The deflection feature of a bimetallic strip is capitalized upon in developing a thermostat. A thermostat is a component of a control system which senses the temperature of the system, and maintains it near a desired set point. In other words, a thermostat is a temperature regulating device.

The thermostat does this by switching a heating or cooling device on or off, and in the process, maintain the desired temperature.

The duration between the switching on and off in order to maintain the temperature to the set point,

is known as hysteresis..

When using a straight Bimetallic strip, an important calculation to determine the movement of the free end of the strip is given by:

Where, d is the deflection, α_A is the coefficient of linear expansion of metal A, α_B is the coefficient of

linear expansion of metal B, T_1 is the lower temperature, T_2 is the elevated temperature, l is the length of the element, and t is the thickness of the element.

The equations show the linear relationship between deflection and temperature change. The operating range of bimetallic elements is from -180 to $+430^{\circ}\text{C}$, and they can be used in application form oven thermometers to home and industrial control thermostats.

However, when the bimetallic element is wound into a spiral, the deflection of the free end of the strip is given by

Where, r is the radius of the spiral, and l is the length of the unwound element.

Example 2.2

A straight bimetallic strip 25cm long and 1.4mm thick is made of copper and invar, and has one end fixed. How much will the free end of a strip deflect if the temperatures change is 180°C ?

Lecture Material: Ohimor E.O.

Page 30

Example 2.3

A 42-cm length of 2.3-mm thick copper-invar bimetallic strip is wound into a spiral with a radius of 5.2cm at 700C. What is the deflection of the strip at 1200C?

(c) Vapour Pressure Thermometer

The vapour-pressure thermometer, as shown in Figure 2.13, consists of a bulb which is partially filled with liquid and vapour, such as methyl chloride, ethyl alcohol, ether, or toluene. In this system, the lowest operation temperature must be above the boiling point of the liquid, and the maximum temperature is limited by the critical temperature of the liquid.

Figure 2.13 Vapour-pressure thermometer

(d) Thermocouples Thermometers

A thermocouple, as shown in Figure 2.14, consists of two dissimilar metals or wires (e.g. iron-constantan) joined at one end known as the hot junction and the other open ends, known as the reference junction, is connected to a device to read the induced current as temperature. It is also referred to as a thermoelectric effect sensor.

Figure 2.14 Thermocouple

Lecture Material: Ohimor E.O.

Page 31

Thermocouple Effects

Three effects are associated with thermocouples:

1. The Seebeck Effect

The seebeck effect states that the voltage produced in a thermocouple is proportional to the temperature between the two junctions, that is

Where, S is the seebeck coefficient, a property of the local material (depends strongly on the composition of the conductor), and ΔT is the gradient in temperature.

2. The Peltier Effect

The peltier effect states that if a current flows through a thermocouple, then one junction is heated (output energy), and the other junction is cooled (absorbed energy), the heat generated in the thermocouple is proportional to the current flowing across the junctions

Where n_A and n_B are the peltier coefficients of conductors A and B respectively, and I is the electric current flowing from A to B.

3. The Thompson Effect

Thompson effect states that when a current flows in a conductor along which there is a temperature difference, the heat produced or absorbed is proportional to the direction of the current and the variation of the temperature.

Where, K is the Thompson coefficient, J is current density and ΔT is temperature gradient.

The Thompson coefficient is related to the seebeck coefficient as

(e) Optical Pyrometers

An optical pyrometer as shown in Figure 2.15, is a temperature measurement instrument that compares the incident radiation from the measured body to the radiation from an internal filament. The current through the filament is adjusted until the radiation colour matches that of the measured body, which is at its peak, glowing a certain shade of red as in the visible spectrum. The current can then be directly related to the temperature of the radiation source.

Page 32

Radiation from Incident Radiation

Figure 2.15 Optical Pyrometer

Measurement inaccuracy of an optical pyrometer is $\pm 5^\circ\text{C}$. The instrument cannot be used in automatic temperature control schemes because the eye of the human operator is an essential part of the measurement system. The reading is also affected by fumes in the sight path. As a result of these difficulties and low accuracy, handheld radiation pyrometers are rapidly overtaking the optical pyrometer in popularity, although the instrument is still widely used in industry for measuring temperatures in furnaces and similar application at present. It is designed to measure temperatures in the range of 5000°C to 10000°C .

(f) Radiation Pyrometers

Radiation pyrometer measure temperature by sensing the heat radiated from a hot body through a fixed lens, and focuses the heat energy onto a thermopile. This is a noncontact device. For instance, furnace temperatures are normally measured through a small hole in the furnace wall. The distance from the source to the pyrometer can be fixed and the radiation should fill the field-of-view of the sensor. Figure 2.16 shows the focusing lens and thermocouple setup, using a thermopile as a detector.

Figure 2.16 Radiation Pyrometer

Lecture Material: Ohimor E.O.

Page 33

2.3 LEVEL, HEIGHT AND VOLUME MEASUREMENT

Level measurement is an important aspect of process control. Level sensing can be single points, continuous direct or indirect. Continuous level monitoring measures the level of the liquid on an uninterrupted basis. In this case, the level of the material will be constantly monitored, and hence the volume can be continuously monitored, if the cross-sectional area of the container is known. This is because, volume, weight and pressure are all directly proportional to depth or height.

#Where, P is the pressure, γ is the specific weight, and h is the depth

There are many widely varying methods for the measurement of liquid level.

Level Measurement

Level sensing devices can be divided into four categories:

- (1) Direct sensing, in which the actual level is monitored;
- (2) Indirect sensing, in which a property of the liquid, such as pressure, is sensed to determine the liquid level;
- (3) Single point measurement, in which it is only necessary to detect the presence or absence of a liquid at a specific level and
- (4) Free flowing solid level sensing.

Direct Level Sensing

A number of techniques are used for direct level sensing, such as sight glass or a float, and ultrasonic distance measuring devices.

The Sight Glass or Gauge

The sight glass or gauge is the simplest method for direct visual reading. The sight glass, usually not longer than 4ft, is normally mounted vertically adjacent to the container. The liquid level then can be observed directly in the sight glass. The ends of the glass are connected to the top and bottom of the tank via shutoff valves. See figure 2.17 below.

Lecture Material: Ohimor E.O.

Page 34

Figure 2.17 Sight glass

Float Sensors

There are two types of floats as shown in Figure 2.18 (a & b); the pulley type and the angular arm type. The float material is less dense than the density of the liquid, and floats up and down on top of the material being measured. In figure 2.18(a), a float with a pulley is used. This method can be used with either liquids or free-flowing solids. With free-flowing solids, agitation is sometimes used to help level the solids. The advantages of the float sensor are that they are almost independent of the density of the liquid or solid being monitored, are accurate and robust, and have a linear output with level height. However, accuracy can be affected by material accumulation on the floor, corrosion, chemical reactions, and friction in the pulleys.

Figure 2.18 Float Sensors

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Page 35